Algorithmic Game Theory

2nd Problem Set

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Problem 1

Solution.

We could add also outcomes such as (1, 0, 0) so 1 of the 3 items are sold, but these outcomes have better options so we dont compute them.

Since the goods are identical, an outcome is a triple (x_1, x_2, x_3) such that

$$x_1 + x_2 + x_3 = 3.$$

The possible outcomes and their corresponding social welfare are shown below:

Outcome (x_1, x_2, x_3)	Social Welfare
$(3, 0, 0)$	2.6
$(0, 3, 0)$	4.1
$(0, 0, 3)$	2.4
$(2, 1, 0)$	4.0
$(2, 0, 1)$	2.9
$(1, 2, 0)$	4.5
$(0, 2, 1)$	4.4
$(1, 0, 2)$	2.7
$(0, 1, 2)$	3.7
$(1, 1, 1)$	3.9

The maximum social welfare is

$$SW^* = 4.5,$$

which is achieved by the allocation

$$(x_1, x_2, x_3) = (1, 2, 0).$$

Therefore, player 1 receives one item, player 2 receives two items, and player 3 receives no item.

VCG Payments

The VCG payment of player i is

$$p_i = \max_{x_{-i}} \sum_{j \neq i} v_j(x_j) - \sum_{j \neq i} v_j(x_j^*).$$

Player 1
Without player 1, the optimal allocation among players 2 and 3 is

$$(0, 2, 1),$$

with welfare

$$v_2(2) + v_3(1) = 3.5 + 0.9 = 4.4.$$

In the optimal allocation of the original auction, the welfare of the other players is

$$v_2(2) + v_3(0) = 3.5 + 0 = 3.5.$$

Hence

$$p_1 = 4.4 - 3.5 = 0.9.$$

Player 2
Without player 2, the optimal allocation among players 1 and 3 is

$$(2, 0, 1),$$

with welfare

$$v_1(2) + v_3(1) = 2 + 0.9 = 2.9.$$

In the optimal allocation of the original auction, the welfare of the other players is

$$v_1(1) + v_3(0) = 1 + 0 = 1.$$

Therefore

$$p_2 = 2.9 - 1 = 1.9.$$

Player 3
Without player 3, the optimal allocation is still

$$(1, 2, 0),$$

with welfare

$$v_1(1) + v_2(2) = 1 + 3.5 = 4.5.$$

In the original allocation, the welfare of the other players is also

$$v_1(1) + v_2(2) = 4.5.$$

Thus

$$p_3 = 4.5 - 4.5 = 0.$$

Final VCG Outcome
Allocation:

$$(x_1, x_2, x_3) = (1, 2, 0).$$

Payments:

$$(p_1, p_2, p_3) = (0.9, 1.9, 0).$$

Utilities:

$$u_1 = v_1(1) - p_1 = 1 - 0.9 = 0.1,$$

$$u_2 = v_2(2) - p_2 = 3.5 - 1.9 = 1.6,$$

$$u_3 = v_3(0) - p_3 = 0.$$

Therefore, the VCG mechanism allocates one item to player 1, two items to player 2, and no items to player 3, while charging payments

$$(0.9, 1.9, 0).$$

Problem 2

Solution.
We show that the social choice function f that selects the path with the largest total cost is not implementable by a truthful mechanism.

A necessary condition for truthfulness is *weak monotonicity*. For every player i, every pair of outcomes a, b, and every pair of valuations v_i, v_i', weak monotonicity requires

$$v_i(a) - v_i(b) \geq v_i'(a) - v_i'(b).$$

Consider the simplest path auction consisting of two parallel edges connecting s to t:

$$a = e_1, \qquad b = e_2.$$

Player 1 owns edge e_1 and player 2 owns edge e_2.

Consider the following two cost profiles.

First profile:

$$c = (1, 0).$$

Since the mechanism selects the path with the largest total cost, it chooses path $a = e_1$. Therefore,

$$f(c) = a.$$

Now consider the second profile:

$$c' = (0, 0).$$

Assume that ties are broken in favor of path $b = e_2$. Hence,

$$f(c') = b.$$

Since we are in a cost domain, we define valuations as the negative of costs:

$$v_i = -c_i.$$

For player 1, under the first profile we have

$$v_1(a) = -1, \qquad v_1(b) = 0.$$

Under the second profile we have

$$v_1'(a) = 0, \qquad v_1'(b) = 0.$$

Applying weak monotonicity gives

$$v_1(a) - v_1(b) \geq v_1'(a) - v_1'(b).$$

Substituting the values:

$$-1 - 0 \geq 0 - 0.$$

Thus,

$$-1 \geq 0,$$

which is impossible.

Therefore, weak monotonicity is violated.

Since weak monotonicity is a necessary condition for truthfulness, there do not exist payment functions

$$p_1, p_2, \dots, p_m$$

such that the mechanism

$$(f, p_1, p_2, \dots, p_m)$$

is incentive compatible.

Hence, the social choice function that selects the path with the largest total cost cannot be implemented by a truthful mechanism.

Problem 3

Solution.

We prove that, in the facility location problem on the interval $[0, 1]$ with single-peaked preferences, a strategyproof social choice function f is onto if and only if it is unanimous.

(Unanimous $\Rightarrow$ Onto)

Assume that f is unanimous. By definition, for every $x \in [0, 1]$,

$$f(x, x, \ldots, x) = x.$$

Therefore, for every point $x \in [0, 1]$, there exists a preference profile whose outcome is x. Hence every point in $[0, 1]$ belongs to the range of f.

Therefore, f is onto.

(Onto $\Rightarrow$ Unanimous)

Assume that f is onto and strategyproof.

Let $x \in [0, 1]$ be arbitrary. Since f is onto, there exists a profile

$$(p_1, p_2, \ldots, p_n)$$

such that

$$f(p_1, p_2, \ldots, p_n) = x.$$

We will show that

$$f(x, x, \ldots, x) = x.$$

Starting from the profile

$$(p_1, p_2, \ldots, p_n),$$

change the players' peaks one by one to x.

First, let player 1 change his peak from p_1 to x. Suppose that after this change the outcome becomes some point $y \neq x$.

Since player 1's true peak is now x, he strictly prefers the outcome x to y because preferences are single-peaked. By reporting p_1 instead of x, the outcome would be x. Hence player 1 could benefit from misreporting, contradicting strategyproofness.

Therefore, the outcome must remain equal to x.

The same argument applies when player 2 changes his peak to x, then player 3, and so on. At each step, strategyproofness implies that the outcome cannot move away from x.

After all players have changed their peaks to x, we obtain the profile

$$(x, x, \ldots, x),$$

and the outcome is still x. Therefore,

$$f(x, x, \ldots, x) = x.$$

Since x was arbitrary, f is unanimous.

We conclude that

$$f \text{ is onto} \quad \iff \quad f \text{ is unanimous.}$$

Problem 4

Solution.

We consider the following instance of the Stable Matching problem with four males and four females.

Men's preferences:

$$\begin{array}{c|l} m_1 & w_1 \succ w_2 \succ w_3 \succ w_4 \\ m_2 & w_2 \succ w_3 \succ w_1 \succ w_4 \\ m_3 & w_3 \succ w_1 \succ w_2 \succ w_4 \\ m_4 & w_1 \succ w_2 \succ w_3 \succ w_4 \end{array}$$

Women's preferences:

$$\begin{array}{c|l} w_1 & m_2 \succ m_3 \succ m_4 \succ m_1 \\ w_2 & m_3 \succ m_4 \succ m_1 \succ m_2 \\ w_3 & m_4 \succ m_1 \succ m_2 \succ m_3 \\ w_4 & m_1 \succ m_2 \succ m_3 \succ m_4 \end{array}$$

We now execute the male-proposing Gale-Shapley algorithm.

Round 1

$$m_1 \to w_1, \qquad m_2 \to w_2, \qquad m_3 \to w_3, \qquad m_4 \to w_1.$$

Woman w_1 prefers m_4 to m_1, therefore she keeps m_4 and rejects m_1.
Current matching:

$$(w_1, m_4), \quad (w_2, m_2), \quad (w_3, m_3).$$

Round 2

$$m_1 \to w_2.$$

Woman w_2 prefers m_1 to m_2, therefore she keeps m_1 and rejects m_2.
Current matching:

$$(w_1, m_4), \quad (w_2, m_1), \quad (w_3, m_3).$$

Round 3

$$m_2 \to w_3.$$

Woman w_3 prefers m_2 to m_3, therefore she keeps m_2 and rejects m_3.
Current matching:

$$(w_1, m_4), \quad (w_2, m_1), \quad (w_3, m_2).$$

Round 4

$$m_3 \to w_1.$$

Woman w_1 prefers m_3 to m_4, therefore she keeps m_3 and rejects m_4. Current matching:

$$(w_1, m_3), \quad (w_2, m_1), \quad (w_3, m_2).$$

Round 5

$$m_4 \to w_2.$$

Woman w_2 prefers m_4 to m_1, therefore she keeps m_4 and rejects m_1. Current matching:

$$(w_1, m_3), \quad (w_2, m_4), \quad (w_3, m_2).$$

Round 6

$$m_1 \to w_3.$$

Woman w_3 prefers m_1 to m_2, therefore she keeps m_1 and rejects m_2. Current matching:

$$(w_1, m_3), \quad (w_2, m_4), \quad (w_3, m_1).$$

Round 7

$$m_2 \to w_1.$$

Woman w_1 prefers m_2 to m_3, therefore she keeps m_2 and rejects m_3. Current matching:

$$(w_1, m_2), \quad (w_2, m_4), \quad (w_3, m_1).$$

Round 8

$$m_3 \to w_2.$$

Woman w_2 prefers m_3 to m_4, therefore she keeps m_3 and rejects m_4. Current matching:

$$(w_1, m_2), \quad (w_2, m_3), \quad (w_3, m_1).$$

Round 9

$$m_4 \to w_3.$$

Woman w_3 prefers m_4 to m_1, therefore she keeps m_4 and rejects m_1.
Current matching:

$$(w_1, m_2), \quad (w_2, m_3), \quad (w_3, m_4).$$

Round 10

$$m_1 \to w_4.$$

Woman w_4 is free and accepts.
Final matching:

$$(w_1, m_2), \quad (w_2, m_3), \quad (w_3, m_4), \quad (w_4, m_1).$$

The algorithm terminates after 10 rounds and makes a total of

$$13$$

proposals.

For n males and n females, the maximum possible number of proposals in the Gale-Shapley algorithm is

$$n^2 - n + 1.$$

For $n = 4$,

$$4^2 - 4 + 1 = 13.$$

Thus, this instance achieves the maximum possible number of proposals.

Problem 5

Solution.

Consider a Vickrey (second-price) auction with one item and two bidders. The valuations of the bidders are independently and uniformly distributed on the interval $[0, 1]$. The reserve price is

$$r = \frac{1}{2}.$$

(i) Expected Revenue

The seller receives revenue only when at least one bidder has value at least r.
There are three possible cases.

Case 1: No bidder has value at least $1/2$.

This means

$$v_1 < \frac{1}{2} \qquad \text{and} \qquad v_2 < \frac{1}{2}.$$

In this case the item is not sold and the revenue is

$$0.$$

The probability of this event is

$$\Pr\left(v_1 < \frac{1}{2},\ v_2 < \frac{1}{2}\right) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Therefore, its contribution to the expected revenue is

$$\frac{1}{4} \cdot 0 = 0.$$

Case 2: Exactly one bidder has value at least $1/2$.
In this case the item is sold at the reserve price $1/2$.
The probability of this event is

$$2 \cdot \Pr\left(v_1 \geq \frac{1}{2}\right) \Pr\left(v_2 < \frac{1}{2}\right) = 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}.$$

Hence the contribution to the expected revenue is

$$\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Case 3: Both bidders have value at least $1/2$.
In this case the item is sold and the winner pays the second-highest bid.
Since there are only two bidders, the payment is equal to

$$\min(v_1, v_2).$$

Conditioned on both bidders having valuations at least $1/2$, we have

$$v_1, v_2 \sim U\left[\frac{1}{2}, 1\right].$$

For two independent uniform random variables on the interval $[a, b]$, the expected value of the minimum is

$$E[\min(X, Y)] = a + \frac{b-a}{3}.$$

Substituting

$$a = \frac{1}{2}, \qquad b = 1,$$

we obtain

$$E[\min(v_1, v_2) \mid v_1, v_2 \geq \tfrac{1}{2}] = \frac{1}{2} + \frac{1/2}{3} = \frac{2}{3}.$$

The probability that both bidders have valuations at least $1/2$ is

$$\Pr\left(v_1 \geq \frac{1}{2},\ v_2 \geq \frac{1}{2}\right) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Therefore, the contribution of this case to the expected revenue is

$$\frac{1}{4} \cdot \frac{2}{3} = \frac{1}{6}.$$

Adding the contributions of all three cases:

$$E[\text{Revenue}] = 0 + \frac{1}{4} + \frac{1}{6} = \frac{5}{12}.$$

Therefore,

$$\boxed{E[\text{Revenue}] = \frac{5}{12}.}$$

(ii) The auction is an affine maximizer
Consider the three possible outcomes:

1. Allocate the item to bidder 1.
2. Allocate the item to bidder 2.
3. Do not sell the item.

Define the affine objective

$$\Phi(o) = \sum_i w_i v_i(o) + c_o,$$

with weights

$$w_1 = w_2 = 1.$$

Choose constants

$$c_{\text{sell}} = 0, \qquad c_{\text{no sale}} = \frac{1}{2}.$$

Then

$$\Phi(\text{bidder 1 wins}) = v_1,$$

$$\Phi(\text{bidder 2 wins}) = v_2,$$

and

$$\Phi(\text{no sale}) = \frac{1}{2}.$$

Hence the mechanism selects the outcome maximizing

$$\max\left\{v_1, v_2, \frac{1}{2}\right\}.$$

If both valuations are below $\frac{1}{2}$, the no-sale outcome is chosen. Otherwise, the item is allocated to the bidder with the highest valuation.

Therefore, the Vickrey auction with reserve price $\frac{1}{2}$ is an affine maximizer.

$$\boxed{\text{The auction is an affine maximizer.}}$$

Problem 6

Solution.

(i)

Consider the game

	L	R
U	$(2,1)$	$(0,0)$
D	$(1,0)$	$(3,1)$

The social welfare of each outcome is the sum of the two players' utilities.

Outcome	Social Welfare
(U,L)	$2+1=3$
(U,R)	$0+0=0$
(D,L)	$1+0=1$
(D,R)	$3+1=4$

Pure Nash Equilibria

For the row player:

- If the column player chooses L, then U gives payoff 2 and D gives payoff 1. Hence the best response is U.
- If the column player chooses R, then U gives payoff 0 and D gives payoff 3. Hence the best response is D.

For the column player:

- If the row player chooses U, then L gives payoff 1 and R gives payoff 0. Hence the best response is L.
- If the row player chooses D, then L gives payoff 0 and R gives payoff 1. Hence the best response is R.

Therefore, the pure Nash equilibria are

$$(U,L)$$

and

$$(D,R).$$

Their social welfare values are

$$SW(U,L)=3,$$

and

$$SW(D,R)=4.$$

The optimal social welfare is

$$OPT = 4.$$

Hence,

$$PoA = \frac{OPT}{\min\{3,4\}} = \frac{4}{3},$$

and

$$PoS = \frac{OPT}{\max\{3,4\}} = \frac{4}{4} = 1.$$

Mixed Nash Equilibrium
Let the row player play U with probability p and the column player play L with probability q. The row player is indifferent when

$$2q = q + 3(1-q).$$

Thus,

$$2q = 3 - 2q$$

and therefore

$$q = \frac{3}{4}.$$

The column player is indifferent when

$$p = 1 - p.$$

Thus,

$$p = \frac{1}{2}.$$

Hence the mixed Nash equilibrium is

$$p = \frac{1}{2}, \qquad q = \frac{3}{4}.$$

Its expected social welfare is

$$\frac{1}{2}\cdot\frac{3}{4}\cdot 3 + \frac{1}{2}\cdot\frac{1}{4}\cdot 0 + \frac{1}{2}\cdot\frac{3}{4}\cdot 1 + \frac{1}{2}\cdot\frac{1}{4}\cdot 4 = 2.$$

Including the mixed equilibrium, the worst equilibrium welfare is 2 and the best equilibrium welfare remains 4.

Therefore,

$$PoA = \frac{4}{2} = 2,$$

and

$$PoS = \frac{4}{4} = 1.$$

(ii)
Consider the following game:

	X	Y	Z
A	$(0.5, 0.5)$	$(0, 0)$	$(0, 0)$
B	$(0, 0)$	$(1, 1)$	$(3, 0)$
C	$(0, 0)$	$(0, 3)$	$(2, 2)$

The pure Nash equilibria are

$$(A, X)$$

and

$$(B, Y).$$

The corresponding social welfare values are

$$SW(A, X) = 1,$$

and

$$SW(B, Y) = 2.$$

The optimal social welfare is attained at

$$(C, Z),$$

with

$$SW(C, Z) = 4.$$

Therefore,

$$PoA = \frac{4}{1} = 4,$$

and

$$PoS = \frac{4}{2} = 2.$$

Thus, the game satisfies

$$\boxed{PoA = 4}$$

and

$$\boxed{PoS = 2}.$$

Problem 7

Solution.

(i)

In the Pigou example, there are two parallel paths from s to t. One path has constant latency 1, and the other path has latency

$$c(x) = x^4.$$

There is one unit of flow.

At Nash equilibrium, all flow uses the path with latency x^4. When $x = 1$,

$$c(1) = 1^4 = 1.$$

Thus,

$$SC(NE) = 1.$$

Now we compute the social optimum. Suppose that x units of flow use the path with latency x^4, and $1 - x$ units use the constant path.

The social cost is

$$SC(x) = x \cdot x^4 + (1 - x) \cdot 1.$$

Therefore,

$$SC(x) = x^5 + 1 - x.$$

To minimize this, we take the derivative:

$$SC'(x) = 5x^4 - 1.$$

Setting this equal to zero gives

$$5x^4 = 1.$$

Hence,

$$x = \left(\frac{1}{5}\right)^{1/4}.$$

The optimal social cost is

$$SC(OPT) = \left(\frac{1}{5}\right)^{5/4} + 1 - \left(\frac{1}{5}\right)^{1/4}.$$

Equivalently,

$$SC(OPT) = 1 - \frac{4}{5}\left(\frac{1}{5}\right)^{1/4}.$$

Thus, the Price of Anarchy is

$$PoA = \frac{SC(NE)}{SC(OPT)} = \frac{1}{1 - \frac{4}{5}\left(\frac{1}{5}\right)^{1/4}}.$$

Therefore,

$$\boxed{PoA = \frac{1}{1 - \frac{4}{5}\left(\frac{1}{5}\right)^{1/4}}}.$$

(ii)
We consider the atomic selfish routing game shown in the figure.
There are four players:

$$1, 2, 3, 4.$$

Their source-target pairs are:

$$s_1 = u, \quad t_1 = v,$$

$$s_2 = u, \quad t_2 = w,$$

$$s_3 = v, \quad t_3 = w,$$

and

$$s_4 = w, \quad t_4 = v.$$

The edges with latency x have cost equal to the number of players using them, while the edges labeled 0 have zero cost.
Consider the following strategy profile:

$$P_1 : u \to w \to v,$$

$$P_2 : u \to v \to w,$$

$$P_3 : v \to u \to w,$$

$$P_4 : w \to u \to v.$$

In this profile, the edge $u \to w$ is used by players 1 and 3, so its cost is 2.
The edge $u \to v$ is used by players 2 and 4, so its cost is also 2.
The edge $w \to v$ is used only by player 1, so its cost is 1.
The edge $v \to w$ is used only by player 2, so its cost is 1.
Therefore, the players' costs are:

$$c_1 = 2 + 1 = 3,$$

$$c_2 = 2 + 1 = 3,$$

$$c_3 = 0 + 2 = 2,$$

and

$$c_4 = 0 + 2 = 2.$$

Hence the social cost of this profile is

$$SC = 3 + 3 + 2 + 2 = 10.$$

We now check that this profile is a Nash equilibrium.

Player 1 could deviate to the direct path $u \to v$. This edge is already used by players 2 and 4, so after the deviation its cost would be 3. Thus player 1 would still have cost 3.

Player 2 could deviate to the direct path $u \to w$. This edge is already used by players 1 and 3, so after the deviation its cost would be 3. Thus player 2 would still have cost 3.

Player 3 could deviate to the direct path $v \to w$. This edge is already used by player 2, so after the deviation its cost would be 2. Thus player 3 would still have cost 2.

Player 4 could deviate to the direct path $w \to v$. This edge is already used by player 1, so after the deviation its cost would be 2. Thus player 4 would still have cost 2.

No player can strictly decrease her cost, so this is a Nash equilibrium.

Now consider the following optimal routing:

$$P_1 : u \to v,$$

$$P_2 : u \to w,$$

$$P_3 : v \to w,$$

$$P_4 : w \to v.$$

Each of the four used edges has exactly one player on it, and therefore each player has cost 1. Thus,

$$SC(OPT) = 4.$$

Therefore,

$$PoA = \frac{SC(NE)}{SC(OPT)} = \frac{10}{4} = \frac{5}{2}.$$

Hence,

$$\boxed{PoA = \frac{5}{2}}.$$

(iii)

We construct a cost-sharing game on an undirected graph with two players.

Let $\delta > 0$ be very small.

Player 1 has two possible paths:

$$P_1 \quad \text{with cost } 2,$$

and

$$Q_1 \quad \text{using one private edge of cost } 1+\delta \text{ and one shared edge of cost } 2.$$

Player 2 also has two possible paths:

$$P_2 \quad \text{with cost } 2-\delta,$$

and

$$Q_2 \quad \text{using only the shared edge of cost } 2.$$

The cost of an edge is shared equally among the players using it.
Consider the profile

$$(P_1, P_2).$$

The social cost is

$$SC(P_1, P_2) = 2 + (2-\delta) = 4-\delta.$$

If player 1 deviates to Q_1, she pays the full cost of the shared edge, since player 2 is not using it. Thus her cost would be

$$1+\delta+2 = 3+\delta > 2.$$

So player 1 does not deviate.
If player 2 deviates to Q_2, she pays the full cost of the shared edge, namely

$$2 > 2-\delta.$$

So player 2 does not deviate either.
Therefore, (P_1, P_2) is a Nash equilibrium.
Now consider the profile

$$(Q_1, Q_2).$$

The shared edge of cost 2 is used by both players, so each player pays 1 for it.
Player 1 also pays her private edge of cost $1+\delta$.
Thus the total social cost is

$$SC(Q_1, Q_2) = 2 + (1+\delta) = 3+\delta.$$

This is the social optimum.
However, (Q_1, Q_2) is not a Nash equilibrium, because player 1 pays

$$1 + (1+\delta) = 2+\delta,$$

and can deviate to P_1 and pay only 2.
Thus, the unique Nash equilibrium is (P_1, P_2), while the optimum is (Q_1, Q_2).
Therefore,

$$PoS = \frac{4-\delta}{3+\delta}.$$

As $\delta \to 0$, this ratio tends to

$$\frac{4}{3}.$$

Hence, for every $\varepsilon > 0$, by choosing $\delta > 0$ sufficiently small, we get

$$PoS \geq \frac{4}{3} - \varepsilon.$$

Thus, there exists a two-player undirected cost-sharing game with

$$\boxed{PoS \geq \frac{4}{3} - \varepsilon}.$$